Dermatology AI for Uganda/East Africa

Research Roadmap & Gap Analysis

Targeting Deployment at Uganda Cancer Institute & Mbarara Hospital

Prepared: September 08, 2026

Status: Pre-Phase 1 — Awaiting Stakeholder Alignment

1. Executive Summary

This roadmap synthesizes a landscape review of 50+ open-source dermatology AI projects, public skin-image datasets, and published fairness/debiasing literature to identify critical gaps for deploying a clinically reliable, fairness-aware dermatology AI system in Uganda (Uganda Cancer Institute, Mbarara Hospital). The analysis reveals that no existing project simultaneously addresses: (a) real African skin-tone data at scale, (b) concept-bottleneck interpretability with clinician-in-the-loop reinforcement learning, (c) operational fairness monitoring in low-bandwidth clinical settings, and (d) integrated web-first deployment with telemetry. This document proposes a phased approach to close these gaps.

2. Landscape Review — Key Findings

2.1 Existing Projects by Category

Category	Representative Projects	Relevance to This Roadmap	Critical Gap
Concept Bottleneck / Transparency	MONET (suinleelab, 89★)	Concept-level explanations grounded in medical literature	No RL/HITL to correct concepts frozen after training
Fairness / Debiasing	CIRCLe (arezou-pakzad, 19★), Detecting-Melanoma-Fairly (shalvamist, 18★), Skin-Deep-Unlearning (pbevan1, 7★)	Bias detection, representation balancing, debiasing during training	Training-time only; no production monitoring for distribution shift
Data Quality / Curation	Corrected-Skin-Image-Datasets (kaumadedunath, 9★), HAM10000 cleaning (ShivamPBhosale, 2★)	Duplicate removal, label error correction	One-time cleaning; no continuous data-quality pipeline
Foundation Models	PanDerm, Derm1M (Siyuan Yan, 74★), DermFM-Zero (24★)	Pre-trained dermatology-specific encoders	Data-centric; no fine-tuning pipeline for African skin tones
Student / Single-Shot Models	LesNet, DermNet crawler, FBQ-weighted ensemble (3★)	Efficient architectures for edge deployment	No fairness guarantees; no HITL; no concept bottleneck; no de

2.2 Identified Loopholes (Gaps in Existing Work)

The following are **not** loops in this project — they are **systemic gaps** across the entire open-source dermatology AI landscape that this roadmap explicitly targets:

1. **Zero real African skin data at scale.** Fitzpatrick17k (17k images) has ~15% Type V-VI; DDI adds ~600 diverse images; ISIC has <5% dark skin. No dataset contains biopsy-confirmed lesions from sub-Saharan African patients (Kaposi sarcoma, mycetoma, tropical dermatoses).

2. **Concept bottleneck models lack reinforcement learning.** MONET provides concept explanations but concepts are fixed post-training. No mechanism for clinicians to correct concepts and propagate updates via RL.

3. **Fairness interventions are training-time only.** CIRCLe, Detecting-Melanoma-Fairly, and Skin-Deep-Unlearning debias during training. None monitor fairness drift in production, especially under distribution shift (rural clinics, mobile capture, different lighting).

4. **Synthetic data bypasses IRB but misses real variance.** GAN-generated images (synthetic-medical-images, 11★) lack rare disease presentations, co-morbidities, and real-world artifacts (motion blur, lighting variation, skin prep differences).

5. **No integrated web-first deployment for low-bandwidth settings.** Most projects are Jupyter notebooks or local scripts. None address: offline-first PWA, image compression for 2G/3G, model quantization for mobile inference, or clinician workflow integration.

6. **HITL is image-level only, not triage-decision level.** Existing human feedback loops collect image labels. None capture the *triage decision* (refer / treat / monitor) with confidence calibration for downstream RL.

7. **Data quality issues propagate silently.** Corrected-Skin-Image-Datasets found duplicates in HAM10000 and Fitzpatrick label errors. No project implements continuous data-quality monitoring for incoming clinical images.

8. **No telemetry from real clinics.** Performance drop on Ugandan phenotypes vs. Western benchmarks is unknown because no system has been deployed with instrumentation.

3. Data Acquisition Strategy

3.1 Primary Sources (Require Partnership & IRB)

Uganda Cancer Institute (UCI): Kaposi sarcoma, HIV-associated dermatoses, biopsy-confirmed labels, Fitzpatrick V-VI, patient demographics, treatment outcomes

Mbarara Hospital (MUST/Mbarara University): General dermatology: eczema, fungal infections, bacterial infections, tropical ulcers, pigmentary disorders, rural patient population

3.2 Secondary Public Datasets (Immediate Use, No IRB)

Dataset	Size	Skin Tone Coverage	Access	Notes
Fitzpatrick17k	17,124	Types I–VI (~15% V-VI)	GitHub/Zenodo	Use Fitzpatrick17k-C (cleaned labels)
DDI (Diverse Dermatology Images)	656 images	Fitzpatrick III–VI focus	Stanford/Zenodo	High-quality clinical + dermoscopy
ISIC 2019/2020	25k+/230k+	<5% Type V-VI	ISIC Archive	Largest; deduplicate via Corrected-Skin-Image-Datasets
HAM10000	10,015	Mostly I–IV	Harvard Dataverse	Known duplicates; use cleaned version
PadUFES-20	2,298	Brazilian population (II–IV)	GitHub	Good Latin American representation
Derm7pt	1,011	Mixed	GitHub	7-point checklist annotations
PH2	200	Mostly I–III	ISIC	High-quality dermoscopy only
DermNet (crawled)	~20k+	Variable	tcxxxx/DermNet-images-crawler	Requires crawling; license check needed

4. Proposed System Architecture

4.1 Core Design Principles

• **Web-first, offline-capable.** PWA with Service Worker; model quantization (INT8/ONNX) for mobile inference; adaptive image compression.

• **Concept bottleneck with RLHF.** Interpretable concepts (ABCD rule, 7-point checklist, pattern analysis) → clinician corrects concepts → reward model updates concept weights via PPO.

• **Fairness as a runtime constraint.** Continuous monitoring of TPR/FPR parity across Fitzpatrick types; automatic alerting + model rollback on drift.

• **Human-in-the-loop at triage decision level.** Clinician selects: Refer / Treat / Monitor + confidence. This (image, concepts, decision, confidence) tuple feeds RL replay buffer.

• **Data quality gates.** Automated duplicate detection (perceptual hashing), Fitzpatrick re-estimation, OOD detection on every upload.

• **Telemetry by default.** Structured logging of: inference latency, concept activations, clinician agreement, fairness metrics, data quality flags — exported to UCI/Mbarara dashboard.

4.2 High-Level Component Diagram

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[Mobile/Web PWA] → [Image Preprocess/Compress] → [Quantized Backbone (EfficientNet/ConvNeXt)] → [Concept Bottleneck Head] → [Triage Policy Head] → [Clinician UI: Concepts + Decision + Confidence] → [RL Replay Buffer] → [Nightly PPO Update] → [Fairness Monitor] → [Model Registry] → [Deploy]
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4.3 Technology Stack

Layer	Technology	Rationale
Frontend	React 18 + TypeScript + Vite + PWA (Workbox)	Offline-first, installable, mobile-optimized
Backend API	FastAPI + Python 3.11 + async SQLAlchemy	High performance, type safety, easy ML integration

ML Runtime	ONNX Runtime Web / PyTorch Mobile + INT8 quantization	Runs in browser (WebAssembly) or native mobile
Concept Bottleneck	Custom head on ConvNeXt-Tiny backbone (6.8M params)	36 separable concepts; small enough for mobile
RLHF	PPO (stable-baselines3) with concept-level reward model	Standard algorithm; concept rewards align with clinician mental model
Fairness Monitor	Custom streaming metrics (TPR/FPR parity, calibration curves)	Real-time, no batch recomputation needed
Data Quality	pHash (imagehash) + CLIP-based OOD detection	Fast duplicate detection; semantic OOD; skin-tone estimation
Database	PostgreSQL (Supabase) + pgvector for embeddings	Managed; vector search for similar cases; HIPAA-ready
Auth	Supabase Auth (OIDC) + role-based access (clinician, researcher, admin)	Standard, well-supported; granular permissions
Telemetry	OpenTelemetry → Grafana Cloud / self-hosted Prometheus	Vendor-neutral; dashboards for UCI/Mbarara
CI/CD	GitHub Actions → Docker → Fly.io / Railway / self-hosted K8s	Automated tests; zero-downtime deploys; rollback on fairness alert

5. Phased Implementation Roadmap

Phase 0: Foundation & Partnerships (Weeks 1–4)

- Execute UCI & Mbarara Data Sharing Agreements + IRB approvals (lead time: 8–12 weeks; start immediately)
- Establish Supabase project, GitHub repo, CI/CD pipeline, staging environment
- Download & preprocess all public datasets (Fitzpatrick17k-C, DDI, ISIC, PadUFES, HAM10000-cleaned)
- Build data quality pipeline: pHash dedup, Fitzpatrick re-labeling (using corrected labels as ground truth), OOD baseline
- Train initial Fitzpatrick skin-type classifier (Types I–VI) on Fitzpatrick17k-C — this becomes the runtime skin-tone estimator
- Define concept vocabulary (ABCD, 7-point checklist, pattern lexicon) with Ugandan dermatologist input

Phase 1: Concept Bottleneck Baseline (Weeks 5–10)

- Train ConvNeXt-Tiny + concept bottleneck head on combined public data (multi-task: diagnosis + concepts)
- Implement concept vocabulary as learnable prototypes; loss = diagnosis CE + concept BCE + orthogonality regularization
- Evaluate on held-out public test sets + fairness slices (Fitzpatrick I–II vs. V–VI)
- Build PWA skeleton: camera capture → compress → ONNX inference → concept visualization → triage decision UI
- Deploy to staging; clinician dry-run with 5 Ugandan dermatologists (remote); collect UX feedback

Phase 2: HITL Data Collection & RLHF Infrastructure (Weeks 11–18)

- Deploy PWA to UCI & Mbarara (pilot: 3 clinicians each site) with IRB-approved data collection protocol
- Collect: (image, concepts, clinician decision [Refer/Treat/Monitor], confidence [1–5], time-to-decision)
- Build RL replay buffer; implement concept-level reward: +1 for correct diagnosis + concept agreement, -1 for disagreement
- Train reward model (Bradley-Terry on pairwise clinician preferences over concept sets)
- Nightly PPO updates on accumulated buffer; fairness gate: only deploy if TPR parity $\Delta < 5\%$ across skin tones
- Implement telemetry dashboards: concept agreement rate, decision latency, fairness metrics, data quality flags

Phase 3: Operational Hardening & Scale (Weeks 19–30)

- Mobile optimization: INT8 quantization, model pruning, progressive image loading, offline queue sync
- Continuous fairness monitoring: streaming TPR/FPR by skin tone, automatic rollback on drift > threshold
- Active learning loop: flag low-confidence / high-disagreement / OOD cases for priority clinician review
- Expand concept vocabulary with Ugandan-specific patterns (Kaposi variants, tropical ulcer stages, mycetoma grains)

• Federated fine-tuning: local updates at each site → secure aggregation → global model (privacy-preserving)

• Clinical validation study design: prospective non-inferiority vs. specialist referral (primary endpoint: sensitivity for malignancy)

Phase 4: Sustainability & Handoff (Weeks 31–40)

• Open-source core platform (MIT/Apache-2); keep Ugandan clinical data under local governance

• Train local ML engineers at MUST/Mbarara on model maintenance, fairness auditing, RLHF pipeline

• Publish: (a) Ugandan skin lesion dataset (de-identified, with consent), (b) fairness-aware concept bottleneck methodology, (c) deployment lessons

• Transition to local ownership: UCI/Mbarara run infrastructure; external team moves to advisory role

6. Risk Register & Mitigations

Risk	Likelihood	Impact	Mitigation
IRB/regulatory delays at UCI/Mbarara	High	High	Start Phase 0 Week 1; parallel track on public data; engage Uganda National C
Insufficient Ugandan data for rare conditions	High	High	Transfer learning from public data + synthetic augmentation (GAN) for rare clas
Clinician adoption resistance	Medium	High	Co-design UI from Day 1; minimal clicks; offline-first; show concept explanation
Fairness drift in production	Medium	High	Streaming fairness monitor with auto-rollback; monthly fairness audit report to
Internet/power instability at clinics	High	Medium	Full offline PWA; background sync; battery-optimized inference; solar charging
Model size too large for mobile	Low	Medium	ConvNeXt-Tiny (6.8M) + INT8 quantization → ~7 MB; test on low-end Android
Concept vocabulary misalignment with local practice	Medium	Medium	Weekly concept review sessions with Ugandan dermatologists in Phase 1–2; v
Funding runway	Medium	High	Phased grants: Phase 0–1 (seed), Phase 2–3 (implementation), Phase 4 (sust

7. Success Metrics (Go/No-Go Gates)

Phase	Metric	Target	Measurement
Phase 1	Diagnosis AUC (public test, all skin tones)	≥ 0.85	Held-out ISIC/DDI/Fitzpatrick17k-C
Phase 1	Concept F1 (dermatologist-annotated concepts)	≥ 0.75	5 clinicians × 200 images
Phase 1	TPR parity (Type I–II vs. V–VI)	$\Delta \leq 0.05$	Stratified evaluation
Phase 2	Clinician concept agreement rate	≥ 70%	Per-session telemetry
Phase 2	RLHF policy improvement vs. supervised baseline	≥ 5% relative AUC gain	A/B on replay buffer
Phase 2	Median decision latency (capture → triage)	≤ 15 seconds	PWA telemetry
Phase 3	Sensitivity for malignancy (prospective)	≥ 90% (non-inferiority margin 5%)	Clinical study vs. specialist
Phase 3	Fairness drift detection time	< 24 hours	Streaming monitor logs
Phase 3	Offline inference success rate	≥ 99%	Field test: 2G, airplane mode
Phase 4	Local team autonomous deployments	≥ 2 per quarter	GitOps logs

8. Open Questions Requiring Stakeholder Decisions

The following must be resolved before Phase 1 kickoff. Each requires explicit agreement from UCI, Mbarara, and the technical team.

Data Governance

- Who owns the Ugandan clinical images and derived models? (Recommendation: UCI/Mbarara retain data sovereignty; model weights open-sourced under Apache-2 with data-use restriction)
- What de-identification standard? (Recommendation: HIPAA Safe Harbor + facial/landmark removal via segmentation)
- Can de-identified data leave Uganda? (Recommendation: No — training runs on-prem or in Uganda-hosted cloud)

Clinical Workflow

- What is the exact triage decision schema? (Options: Refer/Treat/Monitor vs. Urgent/Routine/Reassure vs. custom)
- Who is the ‘clinician’ for HITL? (Dermatologist, clinical officer, nurse? Different expertise levels need different UI)
- Is the AI assistive (clinician decides) or autonomous for certain low-risk classes? (Regulatory implication)

Fairness Definition

- Which parity metric? (TPR parity, FPR parity, equalized odds, calibration parity — recommend equalized odds + calibration)
- What skin-tone grouping? (Fitzpatrick I–II vs. III–IV vs. V–VI, or binary light vs. dark, or Monk Skin Tone Scale)
- What drift threshold triggers rollback? (Recommendation: TPR Δ > 5% for any malignancy class)

Technical Scope

- Target conditions: Start with ‘malignancy vs. benign’ binary, or full multi-class (15+ Ugandan-relevant conditions)?

• Image type: Clinical (phone camera) only, or include dermoscopy? (Dermoscopy requires hardware procurement)

• Language: English only, or Luganda/Runyankore/Rutooro for patient-facing explanations?

Sustainability

• Post-grant infrastructure budget: Who pays for GPU inference, Supabase, monitoring? (Estimate: $500–2000/month)

• Local ML engineering capacity: Does MUST/Mbarara have staff to maintain, or is capacity-building required?

• Publication & IP: Joint authorship policy; patent vs. open-source for novel methods (e.g., concept RLHF)?

Appendix A: Key References

• Abhishek et al., 'Investigating the Quality of DermaMNIST and Fitzpatrick17k Datasets', Nature Scientific Data, 2025. (Fitzpatrick17k-C on Zenodo: 10.5281/zenodo.11101337)

• Groh et al., 'Evaluating Deep Neural Networks Trained on Clinical Images in Dermatology with the Fitzpatrick 17k Dataset', CVPR 2021.

• Daneshjou et al., 'Disparities in Dermatology AI Performance on a Diverse, Curated Clinical Image Set', NEJM AI, 2024. (DDI dataset)

• Yan et al., 'Derm1M: A Million-Scale Vision-Language Dataset Aligned with Clinical Ontology Knowledge for Dermatology', ICCV 2025 Highlight.

• Yan et al., 'DermFM-Zero: Vision-Language Foundation Model for Zero-Shot Dermatology', 2026.

• Pacheco et al., 'CIRCLe: Color Invariant Representation Learning for Unbiased Classification of Skin Lesions', 2023.

• Baldock et al., 'Detecting Melanoma Fairly: Skin Tone Detection and Debiasing for Skin Lesion Classification', MICCAI DART 2022.

• Tschandl et al., 'The HAM10000 Dataset', Scientific Data, 2018.

• Codella et al., 'ISIC Archive', International Skin Imaging Collaboration.

• Perez et al., 'PadUFES-20: A Skin Lesion Dataset from Brazilian Patients', 2020.

Appendix B: GitHub Repositories for Reference Implementation

• https://github.com/kakumarabhishek/Corrected-Skin-Image-Datasets — Data quality pipeline, Fitzpatrick17k-C

• https://github.com/suinleelab/MONET — Concept bottleneck + medical literature grounding

• https://github.com/arezou-pakzad/CIRCLe — Color-invariant fairness

• https://github.com/pbevan1/Detecting-Melanoma-Fairly — Debiasing + skin tone detection

• https://github.com/pbevan1/Skin-Deep-Unlearning — Artifact debiasing

• https://github.com/SiyuanYan1/Derm1M — Million-scale vision-language dataset

• https://github.com/SiyuanYan1/DermFM-Zero — Zero-shot foundation model

• https://github.com/tcxxxx/DermNet-images-crawler — DermNet crawler

• https://github.com/mmu-dermatology-research/isic_duplicate_removal_strategy — ISIC deduplication